

# NLP hw2

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## 1 Word-Level Neural Bi-gram Language Model

(a)

### Step 1: Simplifying the Cross Entropy Function

First, we substitute the definition of the softmax function into the Cross Entropy loss and apply logarithm rules:

$$\begin{aligned}\text{CE}(\mathbf{y}, \hat{\mathbf{y}}) &= - \sum_i y_i \log(\hat{y}_i) \\ &= - \sum_i y_i \log \left( \frac{\exp(\theta_i)}{\sum_j \exp(\theta_j)} \right) \\ &= - \sum_i y_i \left( \log(\exp(\theta_i)) - \log \left( \sum_j \exp(\theta_j) \right) \right)\end{aligned}$$

Since  $\log(\exp(x)) = x$ , we get:

$$= - \sum_i y_i \left( \theta_i - \log \left( \sum_j \exp(\theta_j) \right) \right)$$

### Step 2: Differentiating with respect to $\theta_k$

Now we take the partial derivative with respect to  $\theta_k$ :

$$\begin{aligned}\frac{\partial \text{CE}}{\partial \theta_k} &= - \frac{\partial}{\partial \theta_k} \left[ \sum_i y_i \theta_i - \sum_i y_i \log \left( \sum_j \exp(\theta_j) \right) \right] \\ &= - \left[ y_k \cdot 1 - \sum_i y_i \cdot \frac{\partial}{\partial \theta_k} \log \left( \sum_j \exp(\theta_j) \right) \right]\end{aligned}$$

Using the chain rule for the logarithm term:

$$= - \left[ y_k - \sum_i y_i \cdot \frac{\exp(\theta_k)}{\sum_j \exp(\theta_j)} \right]$$

We can pull out the term that does not depend on  $i$ :

$$= -y_k + \left( \sum_i y_i \right) \frac{\exp(\theta_k)}{\sum_j \exp(\theta_j)}$$

Since  $\mathbf{y}$  is a one-hot vector, the sum of its elements is 1 (i.e.,  $\sum_i y_i = 1$ ). Additionally, by definition,  $\frac{\exp(\theta_k)}{\sum_j \exp(\theta_j)} = \hat{y}_k$ . Substituting these back yields the final result:

$$= -y_k + 1 \cdot \hat{y}_k = \hat{y}_k - y_k$$

(b)

#### Step 1: Applying the Chain Rule

To find the gradient of the loss  $J$  with respect to the input  $\mathbf{x}$ , we use the chain rule. First, let us define the intermediate variables of the network:

$$\begin{aligned} \mathbf{z}_1 &= \mathbf{x}\mathbf{W}_1 + \mathbf{b}_1 \\ \mathbf{h} &= \sigma(\mathbf{z}_1) \\ \mathbf{z}_2 &= \mathbf{h}\mathbf{W}_2 + \mathbf{b}_2 \\ \hat{\mathbf{y}} &= \text{softmax}(\mathbf{z}_2) \end{aligned}$$

The chain rule backward from the loss  $J$  to the input  $\mathbf{x}$  is given by:

$$\frac{\partial J}{\partial \mathbf{x}} = \frac{\partial J}{\partial \mathbf{z}_2} \cdot \frac{\partial \mathbf{z}_2}{\partial \mathbf{h}} \cdot \frac{\partial \mathbf{h}}{\partial \mathbf{z}_1} \cdot \frac{\partial \mathbf{z}_1}{\partial \mathbf{x}}$$

#### Step 2: Calculating Individual Derivatives

Now, we compute each component in the chain separately:

- $\frac{\partial \mathbf{z}_1}{\partial \mathbf{x}} = \mathbf{W}_1^T$  (Derivative of the first linear transformation)
- $\frac{\partial \mathbf{h}}{\partial \mathbf{z}_1}$  represents the derivative of the sigmoid function. It is structured as a  $D_h \times D_h$  diagonal Jacobian matrix where the elements on the main diagonal are the derivatives of each individual activation  $h_i$ , and all off-diagonal elements are zero:

$$\begin{pmatrix} h_1(1-h_1) & 0 & \cdots & 0 \\ 0 & h_2(1-h_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & h_{D_h}(1-h_{D_h}) \end{pmatrix}$$

- $\frac{\partial \mathbf{z}_2}{\partial \mathbf{h}} = \mathbf{W}_2^T$  (Derivative of the second linear transformation)
- $\frac{\partial J}{\partial \mathbf{z}_2} = \hat{\mathbf{y}} - \mathbf{y}$  (This is the gradient of the Cross Entropy loss with respect to the input of the softmax function, which was already derived in section a)

### Step 3: The Final Result

Multiplying all the terms together from left to right, we get the final gradient:

$$\frac{\partial J}{\partial \mathbf{x}} = (\hat{\mathbf{y}} - \mathbf{y}) \mathbf{W}_2^T \begin{pmatrix} h_1(1-h_1) & 0 & \cdots & 0 \\ 0 & h_2(1-h_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & h_{D_h}(1-h_{D_h}) \end{pmatrix} \mathbf{W}_1^T$$

(d)

Based on the training iterations of the neural bi-gram language model using GloVe embeddings, the final evaluation yielded the following result:

- **Dev Perplexity:** 112.8621

*Note: The trained parameters have been saved to `saved_params_40000.npy` and are included in the submission zip file.*

## 2 Generating Shakespeare with a Char-level LM

(a)

Character-based models easily handle out-of-vocabulary words and misspellings by building them character by character, and they require far fewer embedding parameters. Meanwhile, word-based models capture semantic meaning and contextual relationships more naturally. Additionally, word-based models yield shorter sequences, reducing computational costs and making it easier for the network to learn long-range dependencies without forgetting past context.

(c)

## 3 Perplexity

(a)

Let  $x_i = p(s_i | s_1, \dots, s_{i-1})$  represent the conditional probability of token  $s_i$ . We begin with the left-hand side of the perplexity equation:

$$2^{-\frac{1}{M} \sum_{i=1}^M \log_2 x_i}$$

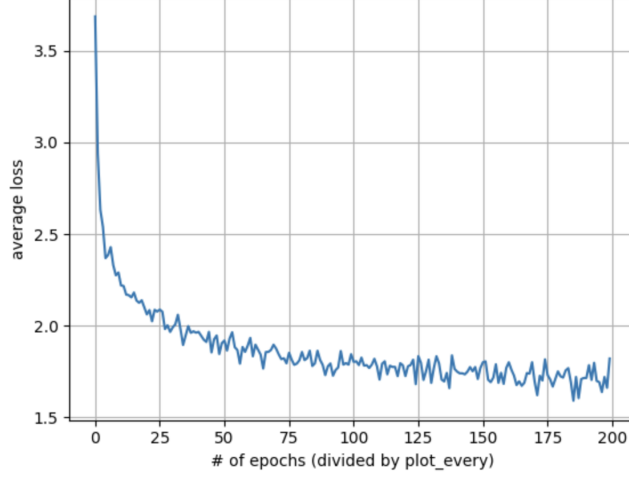


Figure 1: Training loss curve for the character-level Shakespeare language model: average loss versus training-step index, where each tick on the x-axis corresponds to `plot_every` SGD iterations.

Using the algebraic property of exponents where  $a^{B \cdot C} = (a^B)^C$ , we can isolate the summation term from the outer scaling factor:

$$= \left( 2^{\sum_{i=1}^M \log_2 x_i} \right)^{-\frac{1}{M}}$$

Next, using the exponent property where a sum in the exponent translates to a product of bases ( $a^{\sum z_i} = \prod a^{z_i}$ ), we can rewrite the inner expression as:

$$= \left( \prod_{i=1}^M 2^{\log_2 x_i} \right)^{-\frac{1}{M}}$$

Since  $2^{\log_2 x_i} = x_i$ , the terms inside the product simplify directly to the joint probability components:

$$= \left( \prod_{i=1}^M x_i \right)^{-\frac{1}{M}}$$

Next, using the natural logarithm identity, we can express each probability term  $x_i$  in base  $e$  as  $x_i = e^{\ln x_i}$ :

$$= \left( \prod_{i=1}^M e^{\ln x_i} \right)^{-\frac{1}{M}}$$

When multiplying terms with identical bases, we add their exponents together ( $\prod e^{z_i} = e^{\sum z_i}$ ), turning the product back into a summation in the exponent of  $e$ :

$$= \left( e^{\sum_{i=1}^M \ln x_i} \right)^{-\frac{1}{M}}$$

Finally, by applying the power-of-a-power exponent rule once more, we multiply the inner and outer exponents:

$$= e^{-\frac{1}{M} \sum_{i=1}^M \ln x_i}$$

Substituting  $x_i = p(s_i | s_1, \dots, s_{i-1})$  back into the formula gives us the right-hand side of the equation:

$$= e^{-\frac{1}{M} \sum_{i=1}^M \ln p(s_i | s_1, \dots, s_{i-1})}$$

This completes the proof, showing that both expressions are mathematically identical.